

“E-COMMERCE ANALYSIS”

A Preliminary Internship Report submitted to Savitribai Phule Pune University, Pune



In the partial fulfillment for the degree of Bachelor of Engineering in Computer Engineering

Submitted by

Sheetal Mali – PRN No. 72240440K

Under the Guidance of

Miss Komal Suryawanshi Mam



NITS IT Training & Solutions

Academic Year: 2024 - 25
SNJB's Late Sau. Kantabai Bhavarlalji Jain College of Engineering
(Jain Gurukul) Neminagar, Chandwad – 423101, Dist. Nashik, Maharashtra State



ESTD - 1928

Department of Computer Engineering
Certificate

This is to certify that **Ms. Sheetal Mali** have successfully completed this Internship Project and its Report entitled “**E-Commerce Analysis**” under my guidance in partial fulfilment of the requirements for the degree of Bachelor of Engineering in Computer Engineering under the Savitribai Phule Pune University, Pune during the semester 2 of academic year 2024-25.

Date:

Place: Chandwad

Miss Komal Mam

Internship Guide

Miss Mrunal Dahale Mam

Owner and Co-Founder

Dr. R. G. Tated

Principal



Date : 31 Dec 2024

TO WHOM SO EVER IT MAY CONCERN

This is to certify that Miss. Sheetal Mali, a student from SNJB's Late Sau Kantabai Bhavarlalji Jain College of Engineering Chandwad, Nashik, Maharashtra. Sheetal will be undergoing internship program at NetLeap IT Training And Solutions.

She will be undertaking her internship in our Data Analytics Domain. The internship will be of 45 days and shall commence from 31-Dec-2024 in our location.

For any further queries, please contact the undersigned or the HR department.



Director Authority
NetLeap IT Training & Solutions

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Acknowledgement

With deep sense of gratitude, I would like to thank all the people who have lit my path with their kind guidance. I am very grateful to these intellectuals who did their best to help during my internship work.

First of all, I would like to thank top management & all the company people of “**Net Leap IT Training & Solution**” for allowing and guiding me to complete my internship in your esteemed organization. It was an immense pleasure moment for me to work in the company environment.

It is my proud privilege to express deep sense of gratitude to **Dr. R. G. Tated** (Principal) for his comments and kind permission to complete this internship. I remain indebted to **Dr. K. M. Sanghavi** (Head of Computer Engineering Department) for her timely suggestion and valuable guidance.

The special gratitude goes to **Prof. P. N. Achaliya** (Internship Coordinator) and **Prof. Komal Suryawanshi Mam**(Internship Internal Guide) for providing me the valuable guidance in the completion of this internship work. I thank all the colleagues for their appreciable help for my working project during the internship. With various industry owners or lab technicians to help, it has been my endeavor to throughout my work to cover the entire project work.

I am also thankful to our parents for providing their wishful support for the successful completion of the project. And lastly I thank to my all friends and the people who are directly or indirectly involved in my project work during the internship.

Sheetal Manohar Mali

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Internship Company Details

Full Name of Company	-	NITS Training And Solutions
Company Address	-	Floor No.1, Avadhoot Adhar Apartment, Gangapur Rd, above Luthra Agencies, opp. Vihar Misal, Old Gangapur Naka, Signal, Nashik, Maharashtra 422005
Company Website	-	http://www.netleapsolutions.com/
Company Guide Name	-	Miss Komal Suryawanshi Mam
Guide Contact No.	-	
Guide Email Id	-	nits@netleapsolution.com

Student Declaration

I hereby declare that the project entitled “**E-Commerce Analysis** ” is executed as per the course requirement of four-year full time BE program of Savitribai Phule Pune University, Pune. I also declare that this internship report has not been submitted by me or any other person to any other University or Institution for a degree program. This is my own and original work.

Place:Nashik

Date: 29-01-2025

Sheetal Manohar Mali

7224044K

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1. Introduction

E-commerce (electronic commerce) is the activity of electronically buying or selling of products on online services or over the Internet.

Electronic commerce draws on technologies such as mobile commerce, electronic funds transfer, supply chain management, Internet marketing, online transaction processing, electronic data interchange (EDI), inventory management systems, and automated data collection systems. E-commerce is in turn driven by the technological advances of the semiconductor industry, and is the largest sector of the electronics industry.

E-commerce businesses may also employ some or all of the followings:

- Online shopping for retail sales direct to consumers via Web sites and mobile apps, and conversational commerce via live chat, chatbots, and voice assistants;
- Providing or participating in online marketplaces, which process thirdparty business-to-consumer (B2C) or consumer-to-consumer (C2C) sales;
- Business-to-business (B2B) buying and selling;
- Gathering and using demographic data through web contacts and social media;
- Business-to-business (B2B) electronic data interchange; 2
- Marketing to prospective and established customers by e-mail or fax (for example, with newsletters);
- Engaging in pretail for launching new products and services;
- Online financial exchanges for currency exchanges or trading purposes.

E-commerce offers consumers the following advantages:

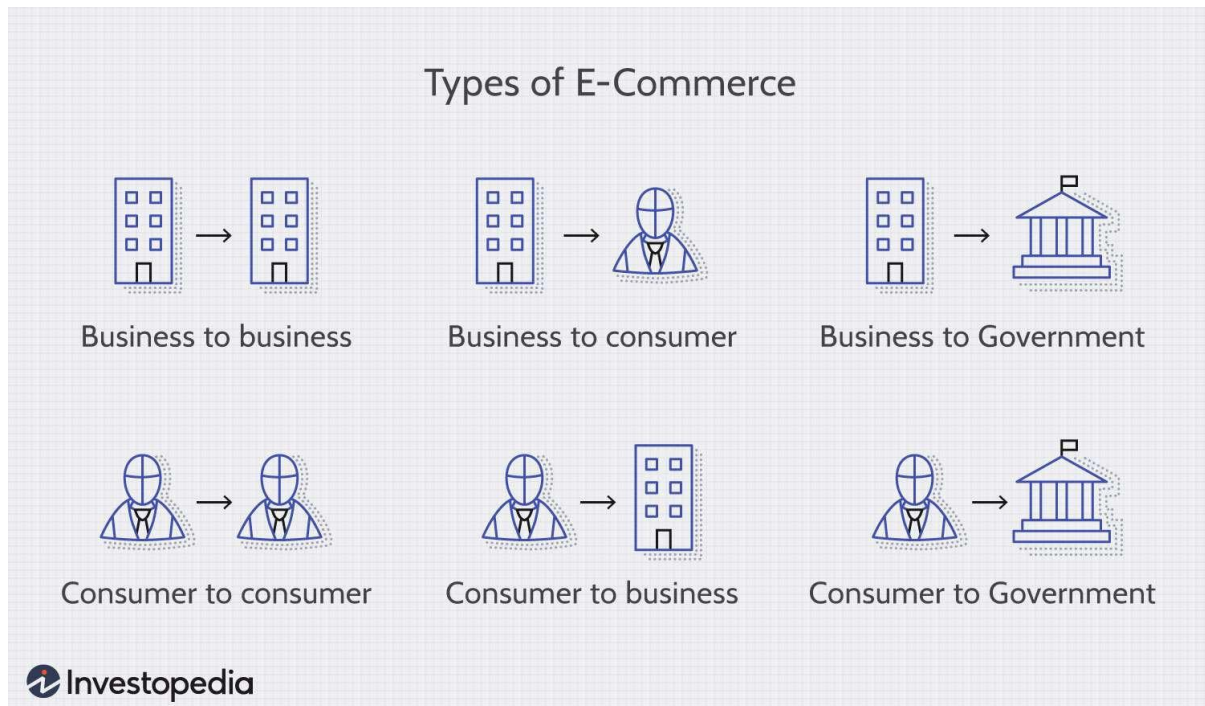
- Convenience. E-commerce can occur 24 hours a day, seven days a week.
- Increased selection. Many stores offer a wider array of products online than they carry in their brick-and-mortar counterparts. And many stores that solely exist online may offer consumers exclusive inventory that is unavailable elsewhere.

E-commerce carries the following disadvantages:

- Limited customer service. If you are shopping online for a computer, you cannot simply ask an employee to demonstrate a particular model's features in person.
- Lack of instant gratification. When you buy an item online, you must wait for it to be shipped to your home or office.

The Indian e-commerce industry has been on an upward growth trajectory. The online retail market in India is estimated to be 25% of the total organized retail market and is expected to reach 37% by 2030. By 2034, it is predicted to surpass the United States to become the second largest e-commerce market globally.

Traditional Types of Ecommerce Business Models:



2. Problem Statement/Objectives of the Internship

The major objectives of the project work on E-commerce industry are: -

- To understand about E-commerce and its role in the economy.
- To understand the types of E-commerce and its functioning
- To understand the influence of the coronavirus pandemic on the E-Commerce Industry with respect to the consumers perspective.
- To understand major academic disciplines contributing to e-commerce research
- To understand about the growth prospects of E-commerce



3. Motivation/Scope of Internship work

Motivation

1. **Market Insights:** Gain a deep understanding of market trends, consumer behavior, and preferences. This can help businesses make data-driven decisions.
2. **Improved Customer Experience:** Analyze customer feedback and purchase patterns to enhance user experience and satisfaction.
3. **Sales Optimization:** Identify best-selling products, seasonal trends, and inventory needs to maximize sales and minimize costs.
4. **Personalized Marketing:** Develop targeted marketing strategies based on customer segmentation and behavior analysis.
5. **Competitive Advantage:** Stay ahead of competitors by leveraging data to understand market dynamics and adjust strategies accordingly.

Scope

1. **Sales Data Analysis:** Analyze historical sales data to identify trends, seasonality, and potential growth areas.
 2. **Customer Segmentation:** Group customers based on purchasing behavior, demographics, and preferences for targeted marketing.
 3. **Inventory Management:** Predict inventory requirements using sales data and optimize stock levels to reduce costs.
 4. **Price Optimization:** Use data to determine optimal pricing strategies to maximize profit while remaining competitive.
- **Website Traffic Analysis:** Examine website traffic data to understand user behavior, improve site navigation, and increase conversion rates.
 - **Sentiment Analysis:** Analyze customer reviews and feedback to gauge sentiment and improve products or services.
 - **Churn Prediction:** Identify factors leading to customer churn and develop strategies to retain customers.
 - **A/B Testing:** Conduct experiments to test different strategies, layouts, or features on the website to see what works best.

4. Methodological details/Live project details

Methodological Details

1. **Data Collection:** Gather data from various sources such as transaction records, customer feedback, website logs, and social media.
2. **Data Cleaning:** Handle missing values, remove duplicates, and correct inconsistencies in the data.
3. **Exploratory Data Analysis (EDA):** Use statistical techniques and visualization tools to understand the data and identify patterns or anomalies.
4. **Feature Engineering:** Create new features from existing data to improve the performance of machine learning models.
5. **Model Building:** Develop predictive models using machine learning algorithms to forecast sales, customer behavior, etc.
6. **Model Evaluation:** Assess the performance of the models using metrics like accuracy, precision, recall, and F1-score.
7. **Deployment:** Implement the models into the e-commerce platform to provide real-time insights and recommendations.
8. **Monitoring and Maintenance:** Continuously monitor the models performance and update them as needed to ensure accuracy and relevance.

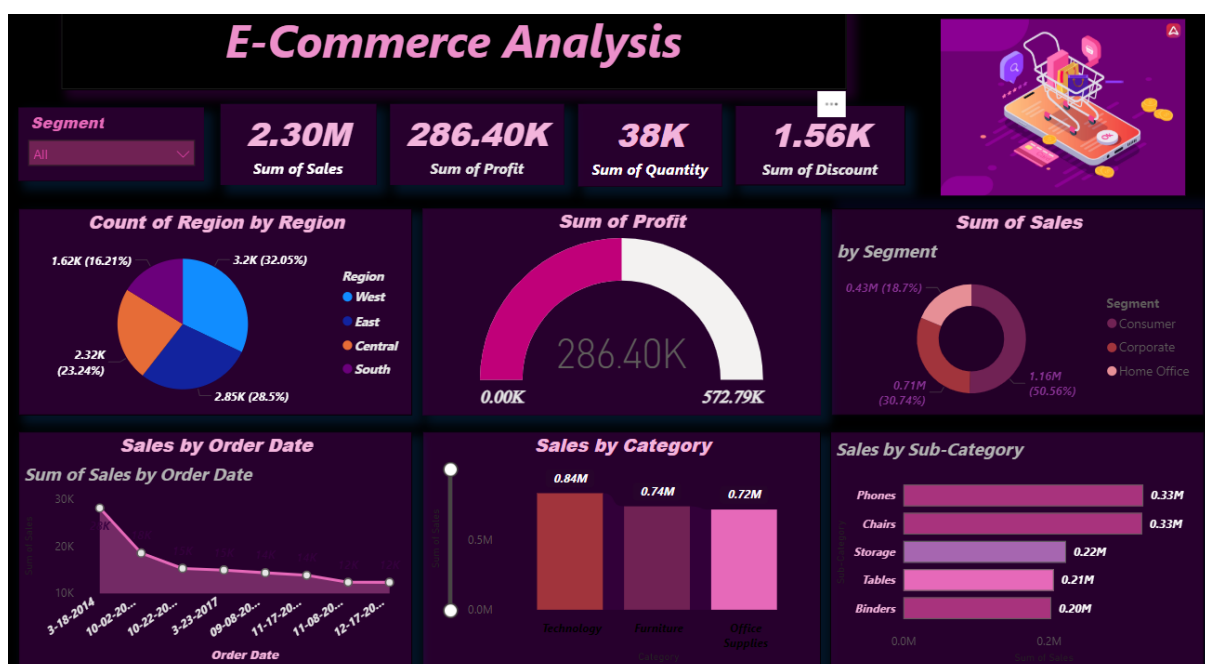
5. Data Analysis & Interpretation of Project

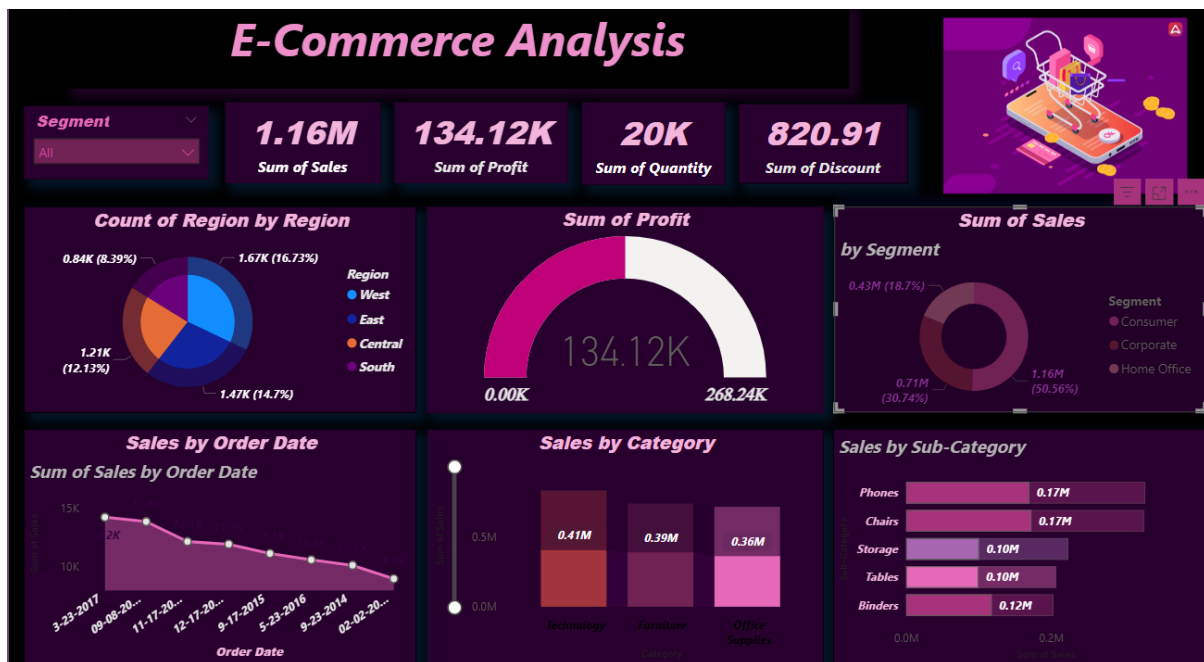
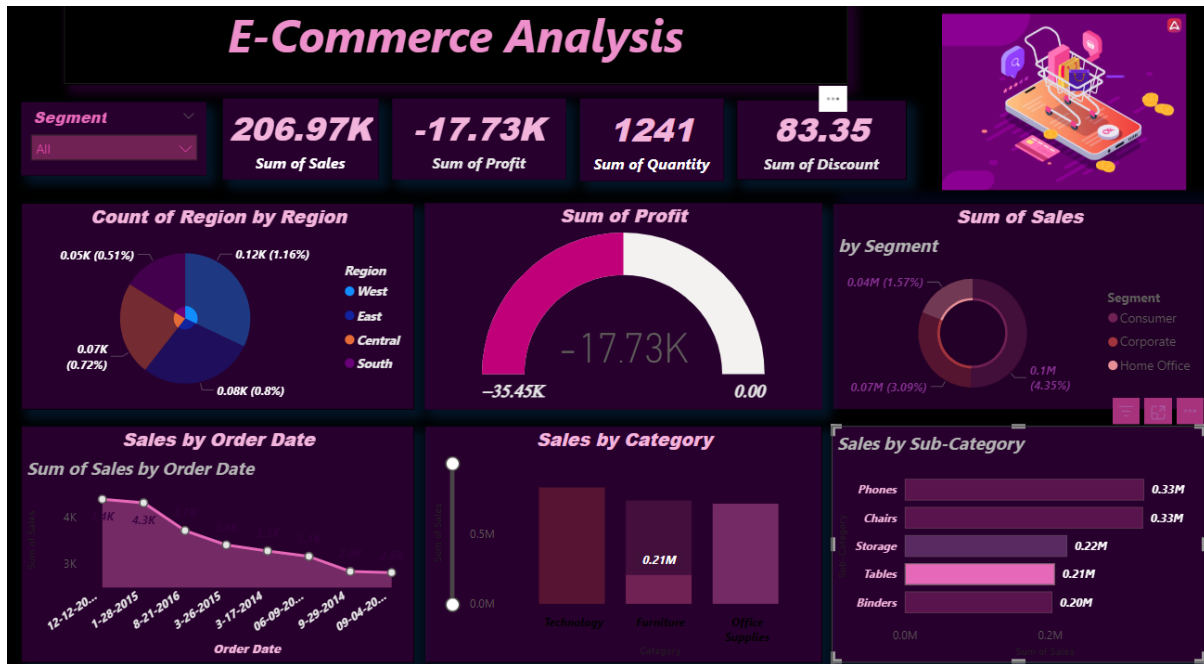
The Covid19 pandemic also known as the coronavirus pandemic emerged in Wuhan of China and was given an account of to the WHO (World Health Organization) on 31st December 2019. On 30th January 2020, Public Health Emergency was announced as a global concern. Covid-19 name was given to this disease on 11th February 2020. This disease plays with the immunity of the human beings, Low immunity individuals are directly affected by this. The pandemic of Covid-19 is a major pitfall not only for the human's health or life or the economic conditions but also on the overall structured society which is not to be changed to an another extend now. Various new things are introduced during this pandemic some of which were their before but still were unknown which now have developed a new scenario. The National Emergency have not spare anyone each and every sector, each and every section is being affected by it whether rich or poor. One of the sectors being affected is the E-Commerce. The Transactions of Buying and Selling that is the business transaction via the route of internet led to the evolution of E-Commerce or broadly as Electronic Commerce. E-Commerce involves dealing with all sort of goods and services over the internet. It is also known as the Internet Commerce because of the way of its functioning online

This paper used a survey of 67 people on their perspective as a consumer involved in online shopping. 9 Questions were asked to them for a clearer picture of the viewpoint changes of consumer during the pandemic of covid-19 towards E-Commerce. The Thought process of the world is now changed with the covid19 pandemic which also bought a change in the lifestyle of people as they are trying to adjust with the new normal and all this led to give a push to E-Commerce industry as well. Covid19 provided E-Commerce with a new environment so as to expand their footprints with quite a few challenges to overcome like productivity or the supply chain effectiveness. Social distancing gave a positive impact on the E-commerce industries growth. The following Table 1 shows the revenue earned by the companies in the Quarter 1 of the previous and current year that is 2019 and 2020 has witness an increase even though the economy is facing a downfall. S.No. Company Revenue 2

6. Result and Learning from Internship

- **Improved Customer Insights:** Gain a deeper understanding of customer behavior, preferences, and trends, enabling more personalized marketing strategies.
- **Enhanced Sales Performance:** Identify best-selling products, peak sales periods, and underperforming items, leading to optimized inventory management and sales strategies.
- **Informed Decision-Making:** Provide data-driven recommendations for product launches, pricing strategies, and promotional campaigns.
- **Increased Revenue:** Implement strategies based on data insights to boost sales and revenue.
- **Optimized Marketing Efforts:** Develop targeted marketing campaigns that resonate with specific customer segments, leading to higher conversion rates.
- **Reduced Customer Churn:** Identify factors leading to customer churn and develop retention strategies to keep customers engaged.
- **Efficient Resource Allocation:** Allocate resources more effectively based on data insights, reducing waste and increasing efficiency.
- **Real-Time Analytics:** Implement real-time data analysis tools to continuously monitor and adapt strategies based on current trends and customer behavior.





7. References

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✓ Ensure that SMEs can participate in e-commerce, for example by providing policy, regulatory or financial incentives for sales diversification and establishing a level playing for SMEs relying on the services of online platforms.

✓ Reduce bottlenecks in the enabling environment for e-commerce, including areas such as connectivity, trade, logistics and postal services.

Thank you!!

